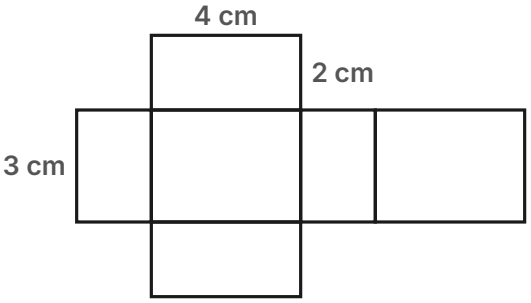
Surface Area and Nets

Open the box flat. Add the areas of all the faces.

A box 4 by 2 by 3, opened flat (a net)



6 faces, in 3 matching pairs.

top and bottom: 2 × 8 = 16

front and back: 2 × 12 = 24

two sides: 2 × 6 = 12

16 + 24 + 12 = 52 sq cm

Surface area is square units (covering). Volume is cubic units (filling).

Your turn!

cube, edge 5 cm: 6 × 25 = 150 sq cm

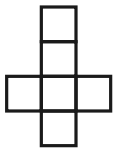
For teachers

Goal: surface area from nets of boxes and square pyramids (6.G.4). Say “Which faces match? Find one, double it.” Surface area is covering (sq units); volume is filling (cubic units).

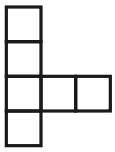
Surface Area and Nets: Practice

Find each face. Matching faces come in pairs.

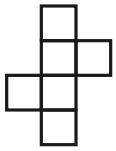
Part 1 Does it fold into a cube? Imagine folding it up. Circle yes or no.

1) 

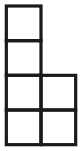
yes no

2) 

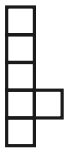
yes **no**

3) 

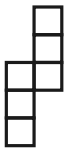
yes no

4) 

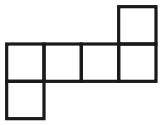
yes **no**

5) 

yes **no**

6) 

yes no

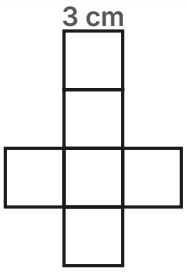
7) 

yes no

8) 

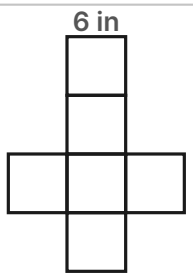
yes **no**

Part 2 Cubes Six faces, all the same.

9) 

one face: 9

6 × 9 = 54 sq cm

10) 

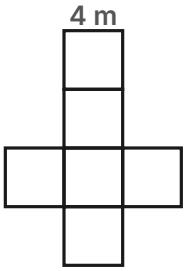
one face: 36

6 × 36 = 216 sq in

Surface Area and Nets: Practice (continued)

Part 2 Cubes (continued)

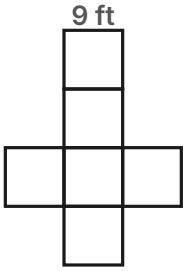
11)



one face: 16

6 × 16 = 96 sq m

12)

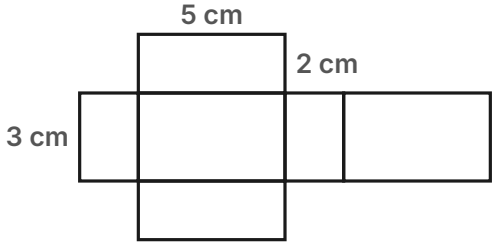


one face: 81

6 × 81 = 486 sq ft

Part 3 Boxes Three pairs of faces.

13)



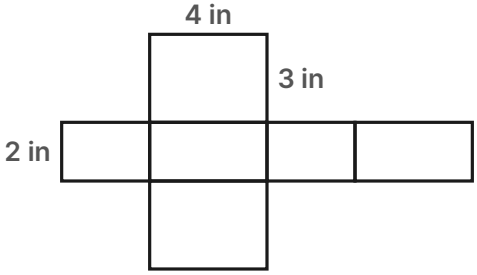
top and bottom: 2 × 10 = 20

front and back: 2 × 15 = 30

two sides: 2 × 6 = 12

surface area = 62 sq cm

14)



top and bottom: 2 × 12 = 24

front and back: 2 × 8 = 16

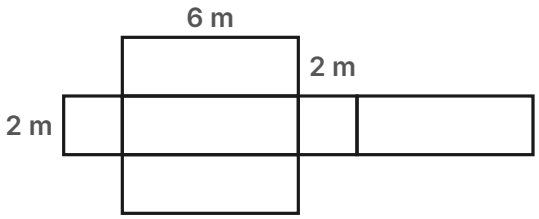
two sides: 2 × 6 = 12

surface area = 52 sq in

Surface Area and Nets: Practice (continued)

Part 3 Boxes (continued)

15)



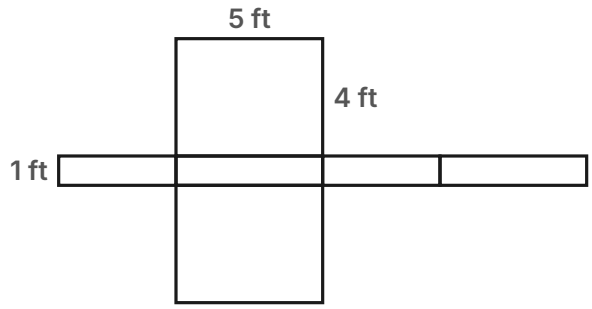
top and bottom: 2 × 12 = 24

front and back: 2 × 12 = 24

two sides: 2 × 4 = 8

surface area = 56 sq m

16)



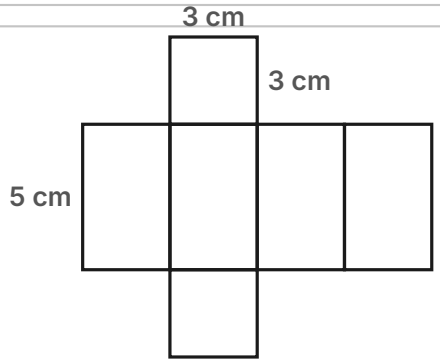
top and bottom: 2 × 20 = 40

front and back: 2 × 5 = 10

two sides: 2 × 4 = 8

surface area = 58 sq ft

17)



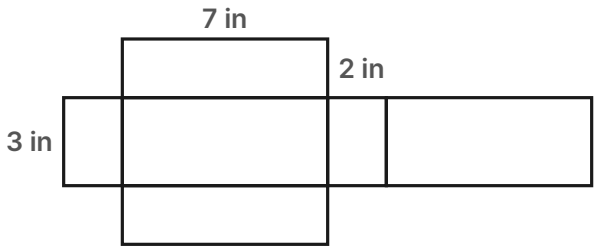
top and bottom: 2 × 9 = 18

front and back: 2 × 15 = 30

two sides: 2 × 15 = 30

surface area = 78 sq cm

18)



top and bottom: 2 × 14 = 28

front and back: 2 × 21 = 42

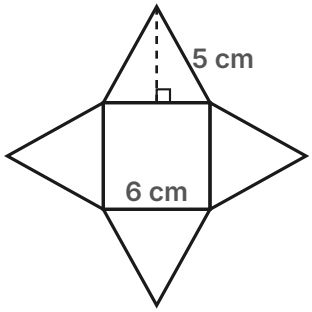
two sides: 2 × 6 = 12

surface area = 82 sq in

Surface Area and Nets: Practice (continued)

Part 4 Square pyramids The square, then 4 triangles.

19)

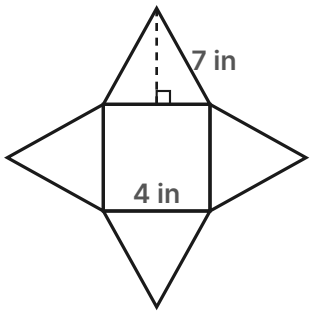


square: 36

4 × 15 = 60

surface area = 96 sq cm

20)

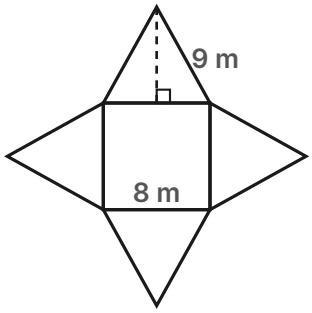


square: 16

4 × 14 = 56

surface area = 72 sq in

21)

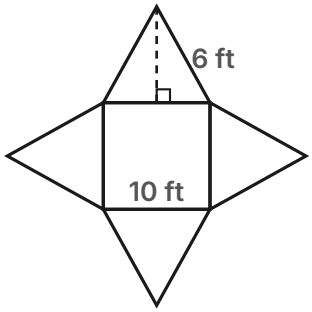


square: 64

4 × 36 = 144

surface area = 208 sq m

22)



square: 100

4 × 30 = 120

surface area = 220 sq ft

Surface Area and Nets: Practice (continued)

Part 5 Boxes No help this time.

23) box 6 × 4 × 3 cm: surface area = 108 sq cm

24) box 5 × 5 × 2 in: surface area = 90 sq in

25) box 8 × 3 × 2 m: surface area = 92 sq m

26) box 7 × 4 × 5 ft: surface area = 166 sq ft

27) box 10 × 2 × 3 cm: surface area = 112 sq cm

28) box 9 × 6 × 1 m: surface area = 138 sq m

29) box 4 × 4 × 7 in: surface area = 144 sq in

30) box 12 × 3 × 2 cm: surface area = 132 sq cm

Part 6 Cubes No help this time.

31) cube, edge 7 cm: surface area = 294 sq cm

32) cube, edge 2 m: surface area = 24 sq m

33) cube, edge 8 in: surface area = 384 sq in

Surface Area and Nets: Practice (continued)

Part 6 Cubes (continued)

34) cube, edge 10 ft: surface area = 600 sq ft

35) cube, edge 11 cm: surface area = 726 sq cm

36) cube, edge 12 in: surface area = 864 sq in

Part 7 More boxes

37) box 12 × 9 × 3 ft: surface area = 342 sq ft

38) box 11 × 7 × 1 cm: surface area = 190 sq cm

39) box 15 × 8 × 8 in: surface area = 608 sq in

40) box 11 × 4 × 8 ft: surface area = 328 sq ft

41) box 13 × 7 × 7 m: surface area = 462 sq m

42) box 10 × 7 × 8 m: surface area = 412 sq m

43) box 7 × 2 × 6 cm: surface area = 136 sq cm

44) box 6 × 2 × 2 ft: surface area = 56 sq ft

Surface Area and Nets: Practice (continued)

Part 7 More boxes (continued)

45) box 12 × 7 × 7 m: surface area = 434 sq m

46) box 13 × 10 × 1 cm: surface area = 306 sq cm

47) box 5 × 3 × 3 cm: surface area = 78 sq cm

48) box 10 × 4 × 6 m: surface area = 248 sq m

49) box 13 × 3 × 1 m: surface area = 110 sq m

50) box 11 × 9 × 5 in: surface area = 398 sq in

51) box 9 × 6 × 2 ft: surface area = 168 sq ft

52) box 14 × 7 × 8 cm: surface area = 532 sq cm

53) box 12 × 5 × 4 cm: surface area = 256 sq cm

54) box 14 × 5 × 2 cm: surface area = 216 sq cm

Surface Area and Nets: Practice (continued)

Part 8 More cubes

55) cube, edge 8 ft: surface area = 384 sq ft

56) cube, edge 12 cm: surface area = 864 sq cm

57) cube, edge 17 cm: surface area = 1,734 sq cm

58) cube, edge 4 ft: surface area = 96 sq ft

59) cube, edge 19 m: surface area = 2,166 sq m

60) cube, edge 20 ft: surface area = 2,400 sq ft

61) cube, edge 14 m: surface area = 1,176 sq m

62) cube, edge 3 ft: surface area = 54 sq ft

Surface Area and Nets: Practice (continued)

Part 9 Remember? Problems from earlier packets.

63) $2\frac{1}{2} \times 2 \times 6 =$ **30**

64) $5x = 85$: $x =$ **17**

65) $2.6 + 13.75 =$ **16.35**

66) -9 **<** -4

67) when $n = 6$: $3n + 11 =$ **29**

68) 18 out of 20 = **90** %

Part 10 Word problems Draw the net. Add the faces.

69) A cube box has edges of 4 in. How much paper covers it all?

Number sentence **6 × 16 = 96**

Answer **96** sq in

Draw or show your work

70) A box is 5 cm by 3 cm by 2 cm. What is its surface area?

Number sentence **30 + 20 + 12 = 62**

Answer **62** sq cm

Draw or show your work

71) A gift box is 10 in by 6 in by 4 in. How much wrapping paper covers it?

Number sentence **120 + 80 + 48 = 248**

Answer **248** sq in

Draw or show your work

Surface Area and Nets: Practice (continued)

Part 10 Word problems (continued)

72) A tent has 4 triangle sides, each with base 3 m and height 4 m. How much cloth for the sides?

Number sentence **3 × 4 ÷ 2 = 6, 4 × 6 = 24**

Answer **24** sq m

Draw or show your work

73) Challenge: A box 6 × 4 × 2 ft has no top. How much wood makes it?

Number sentence **96 − 24 = 72**

Answer **72** sq ft

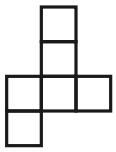
Draw or show your work

Check-In: Surface Area and Nets

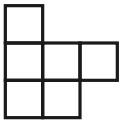
Show what you know. No help this time.

Mastery: 5 of 6 (80%)

Part 1 Does it fold into a cube? Circle yes or no.

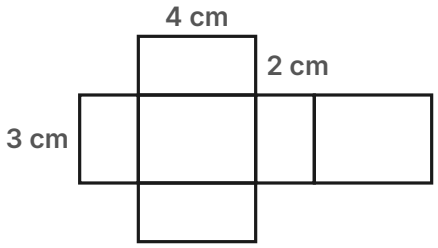
1) 

(yes) no

2) 

yes (no)

Part 2 Boxes

3) 

top and bottom: 2 × 8 = 16

front and back: 2 × 12 = 24

two sides: 2 × 6 = 12

surface area = 52 sq cm

Part 3 Find the surface area

4) cube, edge 5 in: surface area = 150 sq in

5) cube, edge 9 m: surface area = 486 sq m

6) box 8 × 2 × 3 in: surface area = 92 sq in